

The H-A0 hypothesis audit and its analytic removal pathway: a sign-decomposition proof of A-zero uniqueness modulo one anchor inequality

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

H-A0 is one of the two named hypotheses carrying B2's T6 (Proposition A) and one of the four in the Reading-H T6-entry candidate set (unit U1). This note audits exactly what H-A0 asserts, then constructs an ANALYTIC removal pathway: a sign-decomposition proof of the $A = 0$ uniqueness and zero-at-gap structure that replaces the numerical curve-shape certification by closed-form lemmas plus one anchor inequality. Scope: the slaved isotropic family $\{\hat{G}_m\}_{m>0}$ of Math437 v1.2 (the natural $A = 0$ section of the layer; WLOG within the layer per the v1.1 locality remark). No tier action; the GATES.md H-A0 row is unchanged pending operator review of this pathway.

2. What H-A0 asserts (audit)

Per the verbatim GATES.md transcription from Math437 v1.2:

1. **(U) Uniqueness:** the $A = 0$ curve $F_0(m)$ has its unique minimum, value 0 after subtraction, at the gap point m^* (equivalently: $\Delta F_0(m) = F_0(m) - F_0(m^*) \geq 0$ with equality iff $m = m^*$).
2. **(Z) Zero-at-gap:** the minimum value is exactly 0 at the gap point (certified to 10^{-12}), with monotone flanks and $\min \Delta F_0 \geq -10^{-9}$ on a consistent quadrature.
3. **(S) Scheme status:** certification is numerical on a consistent quadrature scheme (internal convergence 3.1×10^{-5}); the 5.5×10^{-3} scheme-gap offset is a recorded measure-convention systematic. The penalty floors P_B are quadrature-free closed forms at the production anchors and do NOT rest on H-A0.

What rests on H-A0 inside the Prop-A proof: exactly the claim-block-C step “ $\Delta F_0 \geq 0, = 0$ iff $m = m^*$ ” that anchors the four-interval decomposition. Nothing else does (the P_B floors, the dip bounds, and the interval boundaries $M_{\pm}, M_c$ are closed-form).

3. The analytic removal pathway (new)

All symbols as in Math437 v1.2: $r = \mu^2 = 0.005$, $u = -0.86$, $v = 3.24$, $M(m) = \int \frac{d^3 k}{(2\pi)^3} [m + C(k^2 - q_0^2)]^{-1}$ (cell-averaged variance of the trial dressing), $M_c = -u/(10v) = 43/1620 = 0.0265432$, $u_{\text{eff}}(M) = u + 10vM$, and the Math437 stationarity identity

$$\frac{dF_0}{dm} = \frac{1}{2} M'(m) g(m), \quad g(m) := r - m + 3u M(m) + 15v M(m)^2,$$

with $g(m^*) = 0$ being EXACTLY the gap equation.

Lemma 1 (L1: strict monotonicity of M). $M'(m) = -\int \frac{d^3k}{(2\pi)^3} [m + C(k^2 - q_0^2)^2]^{-2} < 0$ for every $m > 0$. (Differentiation under the integral: for $m \geq m_0 > 0$ the integrand and its m -derivative are dominated by the integrable m_0 -versions — large- k decay k^{-2} and k^{-6} respectively in the radial measure; this textbook regularity step is the named gap G-A0-DUI below, expected routine.)

Lemma 2 (L2: g is strictly decreasing wherever $u_{\text{eff}} \geq 0$). $g'(m) = -1 + 3u_{\text{eff}}(M(m))M'(m)$. On the region $M(m) \geq M_c$ (equivalently $m \leq m_c$ where $M(m_c) = M_c$): $u_{\text{eff}} \geq 0$ and $M' < 0$ give $g'(m) \leq -1 < 0$.

Lemma 3 (L3: $g < 0$ beyond the window, closed-form). On $M(m) < M_c$ (i.e. $m > m_c$): using $u < 0$ (so $3uM < 0$) and $0 \leq M < M_c$,

$$g(m) < r + 15vM_c^2 - m = 0.005 + 48.6 \times (43/1620)^2 - m = 0.039241 - m.$$

Hence $g(m) < 0$ for every $m > m_w := 0.039241$.

Theorem 1 (A-zero uniqueness and zero-at-gap, conditional skeleton). Assume (i) L1 (modulo G-A0-DUI), and (ii) the ANCHOR WINDOW INEQUALITY $m^* > m_w$. Then F_0 is strictly decreasing on $(0, m^*)$ and strictly increasing on (m^*, ∞) ; consequently $\Delta F_0(m) > 0$ for all $m \neq m^*$ and $\Delta F_0(m^*) = 0$ — i.e. H-A0's (U) and (Z) hold as a THEOREM, with no curve-shape numerics.

Proof. Since $M(m^*) = M_R > M_c$ (the recorded anchor inequality, $\times 4.1$ margin), $m^* < m_c$. On $(0, m_c]$: L2 gives g strictly decreasing; $g(m^*) = 0$ forces $g > 0$ on $(0, m^*)$ and $g < 0$ on $(m^*, m_c]$, and in particular m^* is the UNIQUE zero of g on $(0, m_c]$. On (m_c, ∞) : $m > m_c \geq m^* > m_w$ — wait, the chain needed is $m > m_c$ and $m_c > m_w$; the latter follows from $m_c > m^* > m_w$ (window inequality) — so L3 gives $g < 0$ there; no further zero. By L1, $\text{sign}(dF_0/dm) = -\text{sign}(g)$: $F'_0 < 0$ on $(0, m^*)$ and $F'_0 > 0$ on (m^*, ∞) . Strict global minimum at m^* , value 0 after subtraction, monotone flanks. $\square$

The anchor window inequality (closed-form arithmetic, $\times 7.8$). The gap equation at the production anchors gives

$$m^* = r + 3uM_R + 15vM_R^2 = 0.005 - 2.58 \times 0.109414 + 48.6 \times 0.109414^2 = 0.304523,$$

which reproduces the certified artefact anchor $r_R = 0.3045257$ to 5×10^{-6} (the residual is M_R display rounding) — so $m^* = r_R$ identically by the gap equation, and

$$\frac{m^*}{m_w} = \frac{0.30453}{0.039241} = 7.76.$$

The window inequality holds with $\times 7.8$ closed-form margin at the production anchors.

4. What remains of H-A0 after the reduction

- **Removed from the load-bearing chain:** the numerical curve-shape certification (monotone flanks, $\min \geq -10^{-9}$, zero-at-gap to 10^{-12}) and with it the SCHEME-DEPENDENCE of the uniqueness statement — the theorem above holds for the exact continuum $M(m)$ and verbatim for any fixed quadrature-consistent discretization of it (the sign decomposition is structural).

- **Demoted to dependency (not hypothesis)**: the anchor values (r_R, M_R) and the inequality $M_R > M_c$ ($\times 4.1$, already recorded quadrature-free at the anchors). The 5.5×10^{-3} scheme-gap offset then concerns only the PROVENANCE of the anchor calibration (shared with the entire production chain via A1-KERNEL-CONV), not the uniqueness/zero-at-gap structure.
- **Named residual gaps**: G-A0-DUI — differentiation-under-integral regularity for $M(m)$ (textbook dominated convergence; expected routine); G-A0-VER — machine verification of the two new arithmetic identities ($m_w = r + 15vM_c^2 = 0.039241$ exact; $m^* = r + 3uM_R + 15vM_R^2 = r_R$ to anchor precision) by a small self-test script (registered follow-up; the agent sandbox shell was unavailable this session, so the asserts are NOT yet executed — the numbers above are exact closed-form arithmetic on certified anchor constants, flagged accordingly).

Upon closure of G-A0-DUI + G-A0-VER and operator review, the GATES.md H-A0 row can be replaced by the weaker, sharper hypothesis “H-ANCHOR: the production anchor pair (r_R, M_R) with $M_R > M_c$ ” — or absorbed entirely into the A1-KERNEL-CONV dependency — which would shrink the Reading-H T6 hypothesis set of unit U1 from four named hypotheses to three plus a convention dependency.

5. Numerical verification

No new machine numbers are executed by this note (sandbox shell unavailable). The arithmetic content is two exact closed-form identities on already-certified constants:

$$m_w = 0.005 + 48.6 (43/1620)^2 = 0.0392414,$$

$$m^* = 0.005 - 2.58 M_R + 48.6 M_R^2|_{M_R=0.109414} = 0.304523,$$

to be machine-asserted by the registered G-A0-VER follow-up script (planned: [codes/vacuum/ha0_sign_decomposition.py](#) with self-test asserts + JSON artefact under claims/B2-PROP-HLAYER/runs/). Existing certified inputs: $M_R = 0.109414$, $M_c = 43/1620$, $r_R = 0.3045257$ (artefact claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json, field anchor; Math437 v1.2 closed forms).

Quantitative sanity check (mandatory): dimensional — g and m carry the same (mass²-like) units as r ; m_w and m^* are directly comparable; magnitude — $m^*/m_w = 7.76$ and the recorded $M_R/M_c = 4.12$ are independent margins and both exceed 4; limit case — as $r \rightarrow 3u^2/(20v)$ (PD branch merge) $M_{\pm}$ coincide and the band disappears, while L3’s window constant $r + 15vM_c^2 \rightarrow 3u^2/(20v) + 3u^2/(20v) \dots$ stays finite and the decomposition degenerates gracefully (the theorem’s interval structure collapses before its inequalities fail); sign-direction — dropping $3uM < 0$ in L3 is one-sided in the SAFE direction (enlarges g , shrinks the claimed region), and $u_{\text{eff}} \geq 0 \Rightarrow 3u_{\text{eff}}M' \leq 0$ likewise only strengthens $g' \leq -1$.

6. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “The stationarity identity $dF_0/dm = \frac{1}{2}M'(m)g(m)$ is quoted from Math437, not re-derived here; if it carries hidden quadrature dependence the whole pathway inherits it.” VALID-with-mitigation: the identity is the variational chain rule of the slaved family (F_0 depends on m only through the trial dressing, and the envelope derivative collapses to the explicit m -dependence), recorded in the certified Math437 v1.2 with the gap-equation zero verified exactly. The G-A0-VER follow-up will re-derive it symbolically alongside the arithmetic asserts; until then the pathway is honestly T3.

- (β) “*L3’s window uses $M < M_c$ but bounds $15vM^2$ by $15vM_c^2$ — if M could exceed M_c on $m > m_c$ the bound fails.*” DISMISSED: $m > m_c \Rightarrow M(m) < M(m_c) = M_c$ is exactly L1’s strict monotonicity (the region is DEFINED by the inequality); and $M \geq 0$ holds because M is an integral of a positive integrand. Both directions of the squeeze are structural.
- (γ) “*The proof needs $m^* < m_c$ AND $m_c > m_w$; the text derives the second from $m_c > m^* > m_w$ — but is $m^* < m_c$ itself certified?*” DISMISSED with the recorded anchor inequality: $M(m^*) = M_R = 0.109414 > M_c = 0.0265432$ ($\times 4.1$, quadrature-free at the anchors per the H-A0 row itself), and L1’s monotonicity converts it to $m^* < m_c$. The chain $m_c > m^* > m_w$ then closes with the $\times 7.8$ window margin.
- (δ) “*This note could be read as discharging H-A0 already.*” CORRECT concern, and the note does NOT do that: the status is T3 PROOF SKETCH with two named residual gaps (G-A0-DUI textbook regularity; G-A0-VER machine asserts not yet executed). The H-A0 row, the B2 hypothesis field, and the U1 candidate set are all UNCHANGED; the replacement proposal (H-A0 $\rightarrow$ H-ANCHOR or absorption into A1-KERNEL-CONV) is for operator review after the gaps close.

7. Result footer

Result ID:	B2-PROPA-HLAYER / ha0-removal-pathway (unit U2)
Precise statement:	REMOVAL PATHWAY (T3 sketch): H-A0’s uniqueness + zero-at-gap reduce to L1 ($M' < 0$; textbook regularity gap G-A0-DUI) + L2 ($g' \leq -1$ on $u_{\text{eff}} \geq 0$) + L3 ($g < 0$ for $m > m_w = 0.039241$, closed form) + the anchor window inequality $m^* = r_R = 0.30453 > m_w$ ($\times 7.8$, closed-form arithmetic on certified anchors). The curve-shape numerics and the scheme dependence of the uniqueness statement become removable; what remains is anchor-value provenance (dependency, not hypothesis).
Scope:	Slaved isotropic family (A=0 section of the layer, WLOG within the layer per Math437 v1.1 locality remark); production operating point.
Dependencies:	Math437 v1.2 stationarity identity + closed forms (certified, Math442); anchor constants (A1-KERNEL-CONV provenance); B5 artefact anchor fields (r_R, M_R).
Evidence grade:	ANALYTIC SKELETON (T3) -- closed-form arithmetic on certified constants; machine asserts NOT yet executed (G-A0-VER registered).
Reproduction command:	(registered follow-up) python codes/vacuum/ha0_sign_decomposition.py
Expected output:	asserts: $m_w = 0.0392414$ (exact rational arithmetic); m^* identity to $1e-5$ of r_R ; $m^*/m_w > 7$; $M_R/M_c > 4$; JSON artefact under claims/B2-PROPA-HLAYER/runs/.

Falsification gate: $m^* \leq m_w$ at the anchors (would void the window); a demonstration that the stationarity identity fails for the slaved family (would void the pathway); G-A0-DUI failing (would be a surprise: textbook dominated convergence).

Tier before / after: B2 unchanged at T6 on {H-LAYER, H-A0}; this note enters at T3. PROPOSED upon gap closure + operator review: H-A0 $\rightarrow$ H-ANCHOR (weaker) or absorption into A1-KERNEL-CONV.

No-overclaim statement: H-A0 is NOT discharged by this note. The pathway is a T3 skeleton with two named gaps; no row, tier, or hypothesis field changes. PDF build + linter execution PENDING operator-side (sandbox shell unavailable to this agent session).

Next required action: G-A0-VER script (machine asserts + JSON); G-A0-DUI one-paragraph regularity writeup; then operator review of the H-A0 $\rightarrow$ H-ANCHOR replacement. Mainline continues: H-LAYER residual inventory (unit U3).